Understanding Model Flatness in Reinforcement Learning and Generative Models

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1. Introduction: Loss Landscape in Deep Models
2. Reinforcement Learning Perspective (ICLR'25)
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Proxima Intelligence (π) Lab @ UNIST



PI: Dr. Sung Whan Yoon
2 Postdocs, 9 PhD students, 10 MS students

AI Robustness, Theory, Next-Generation Intelligence

AI4X/Sci AI-based Applications (e.g., EEG, Industrial AI, Quantum)

AI+Comm/Net Intelligent Communications

Artificial Intelligence: Robustness, Generalization

[Multimodal Theory] ICML'26, ICLR'25
[Robust RL] ICLR'25
[Generalization] AAAI'26, AISTATS'24
[Generative Robustness] ICCV'25
[Quantum ML] ICML'26, CIKM'26
[Trustworthy] ICML'26
[FL] CVPR'26, ECCV'26 ICML'24, WACV'24

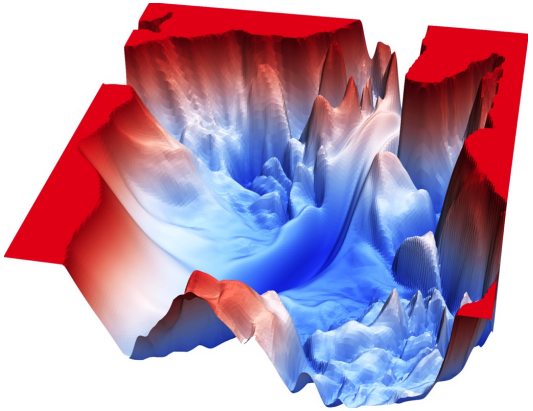
AI4X & AI4Sci

[Industrial AI] (project) RL Gas Combustion
[EEG] J-NER'25, EAAI'26 (submitted),
(ongoing) EEG-Multimodal
[VLA] (Ongoing) Robust VLA
[AI4Sci] (Ongoing) QML Robustness/Trustworhty

AI+Comm./Net.

[6G] JSAC'25, WoWMoM'24, CML'26

Preliminaries: Flat Minima Searching



Loss values on the space of model parameters: $\mathcal{L}(\mathcal{D}; w)$.

Training is to find the valley with the **minimal loss value via SGD**.

Flat minima is the minima with **a flatter shape of loss landscape**.

Figure 1: Loss landscape (conceptual visualization)

Preliminaries: Flat Minima Searching

What is the meaning of flatness (or sharpness) minima?

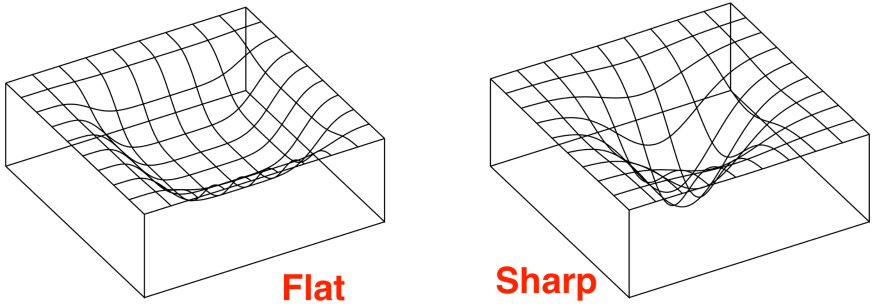


Figure 2: Flat vs. sharp minima

When the loss slowly varies around the minima, we can call the minima ***flat minima***.
Otherwise, when the loss value rapidly changes around minima, which means ***sharp minima***.

Preliminaries: Flat Minima Searching

In the **classification task (supervised settings)**,
flatness links to **model robustness**
against the **distribution shift** from training to testing.

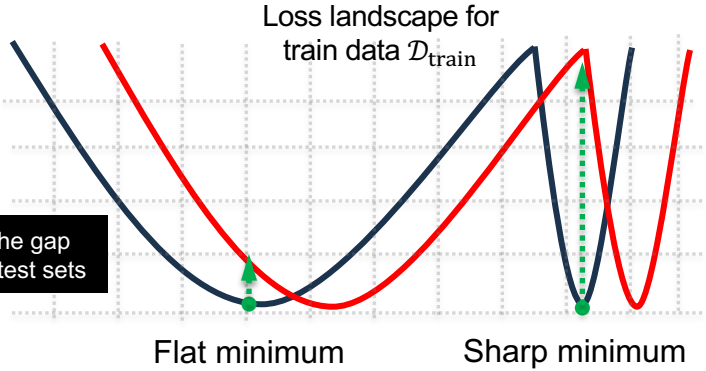


Figure 3: Flat vs. sharp minima given a data distribution shift

Preliminaries: Flat Minima Searching

Authors of [SAM'21] proposed Sharpness-Aware Minimization (SAM)
that adopts the sharpness term in the optimization objective of deep models.

$$\left[\max_{\|\epsilon\|_2 \le \rho} L_{\mathcal{S}}(\boldsymbol{w}+\boldsymbol{\epsilon}) - L_{\mathcal{S}}(\boldsymbol{w})\right] + L_{\mathcal{S}}(\boldsymbol{w}) + h(\|\boldsymbol{w}\|_2^2/\rho^2)$$

Sharpness (or flatness) — Loss at minima — L2 regularization

Eq. 1: SAM objectives

Even with the model perturbations, the loss should be stabilized to flatter region.

[SAM'21] P. Foret et al., "SHARPNESS-AWARE MINIMIZATION FOR EFFICIENTLY IMPROVING GENERALIZATION," ICLR 2021.

Preliminaries: Flat Minima Searching

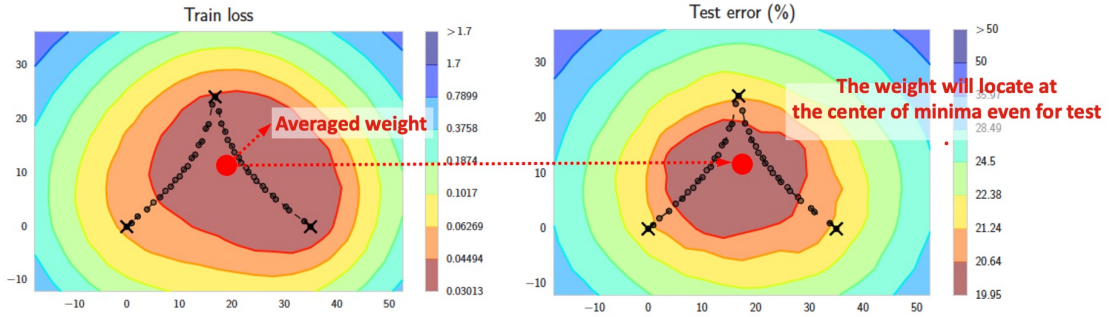


Figure 4: Stochastic Weight Averaging

As another approach, weight averaging promotes flatter minima.

[UAI'18] P. Izmailov et al., "Averaging Weights Leads to Wider Optima and Better Generalization," UAI 2018.

Preliminaries: Flat Minima Searching

Its benefits are broadly observed in various perspectives
, including **domain generalization** [SWAD'21] and **adversarial robustness** [AT'24].

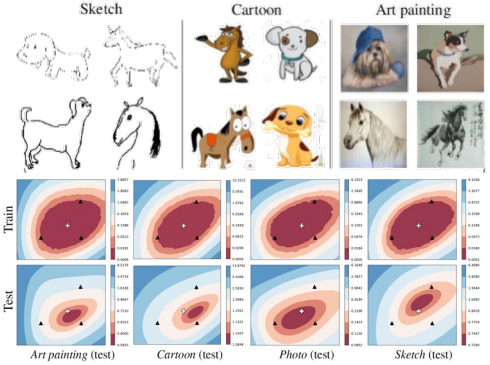


Figure 5: Flat minima in domain generalization [SWAD'21]

$$\mathbb{E}_{(\boldsymbol{x},y)\sim\mathcal{D}}\left[\max_{||\boldsymbol{\delta}||_\infty\leq\epsilon}\mathcal{L}(\boldsymbol{x}+\boldsymbol{\delta},y;w)\right]$$

Adversarial training

$$\mathbb{E}_{(x,y)\sim\mathcal{D}}\left[\max_{|\delta|\leq\epsilon}\mathcal{L}(x,y;w+\delta)\right]$$

SAM

Theorem 4.6. *Denote the perturbation range ϵ of AT and SAM as ϵ_{AT} and ϵ_{SAM}, respectively. Then, when both methods return the same robust feature weight W_R, we have the following relation between ϵ_{AT} and ϵ_{SAM}:*

$$2+\frac{3}{\epsilon_{SAM}^2}\approx\frac{2\eta}{\epsilon_{AT}} \tag{15}$$

Eq. 2: SAM-AT duality [AT'24]

[SWAD'21] J. Cha et al., "SWAD: Domain Generalization by Seeking Flat Minima," NeurIPS 2021.
[AT'24] Y. Zhang et al., "On the Duality Between Sharpness-Aware Minimization and Adversarial Training," ICML 2024

Preliminaries: Flat Minima Searching

Advantages of flat minima are well-documented in supervised training:

Classification [SAM'21, SWA'17], Distributed optimization [FedSAM'22, FedGF'24],
Domain Generalization [SWAD'21], Adversarial training [AT'24]

**How about other learning frameworks,
including generative modeling and reinforcement learning?**

**Can we naively extend the flatness robustness
to the different learning frameworks? *NO***

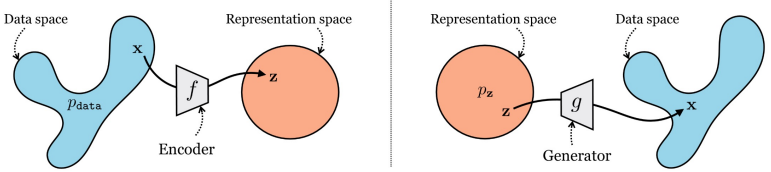
[FedSAM'22] Z. Qu et al., "Generalized Federated Learning via Sharpness Aware Minimization," ICML 2022.
[FedGF'24] T. Lee and S. W. Yoon, "Rethinking the Flat Minima Searching in Federated Learning," ICML 2024

Preliminaries: Our Picture

Deep generative model:

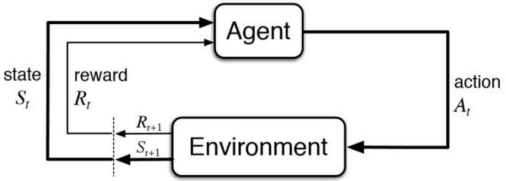
$$\mathcal{L}_{\text{SGM}} = \mathbb{E}_t \left[\lambda(t) \cdot \mathbb{E}_{p_t(\mathbf{x})} \left[\| s_\theta(\mathbf{x}, t) - \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}) \|_2^2 \right] \right]$$

(a score-based objective, e.g., diffusion models)



Deep reinforcement learning:

$$\nabla_\theta V^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t \nabla_\theta \log \pi_\theta(a_t|s_t) R_t \right]$$



ICLR International Conference On Learning Representations

"Reinforcement Learning Perspective"

MIIT Machine Intelligence and Information Theory Laboratory | UNIST ULSAN NATIONAL INSTITUTE OF SCIENCE AND TECHNOLOGY

Published as an ***Oral presentation*** in ICLR'25 (1.7% of submissions)

Flat Reward in Policy Parameter Space Implies Robust Reinforcement Learning



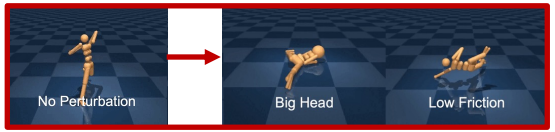
Hyun Kyu Lee
(MIIT Lab & Korea Navy)

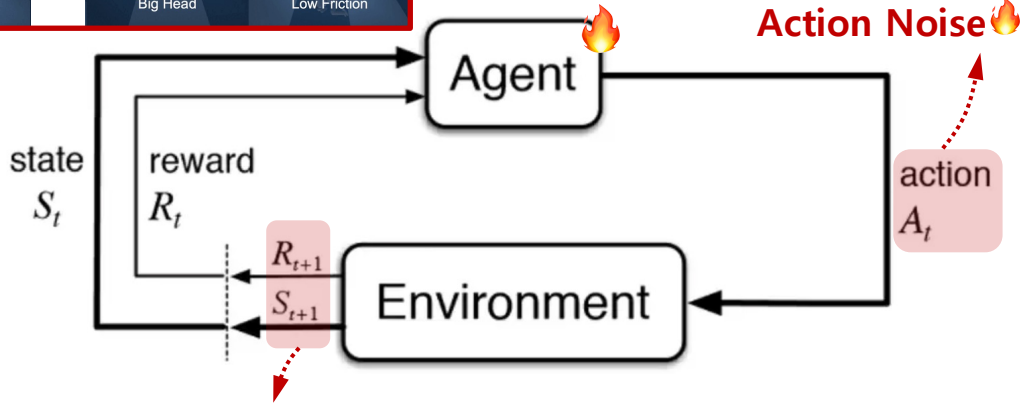


Sung Whan Yoon†
(MIIT Lab)

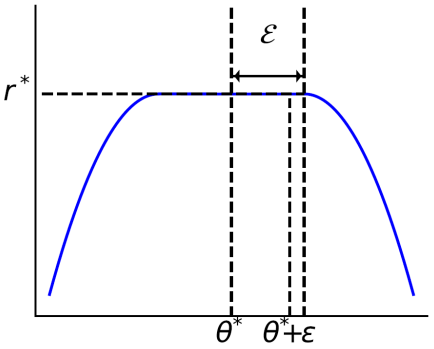
†Corresponding author

Robust Reinforcement Learning



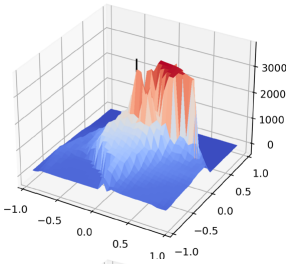


Reward Flatness

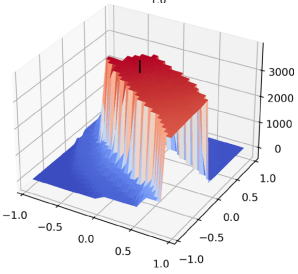


$\varepsilon \downarrow$

$\varepsilon \uparrow$



"Sharpness"

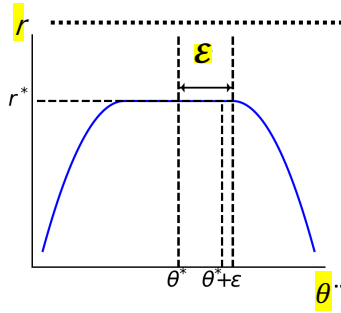


"***Flatness***"

Reward Flatness

Ensuring stability of expected cumulative reward under parameter perturbation

- Flat $\boldsymbol{\varepsilon}$
- Reward r
- Policy θ

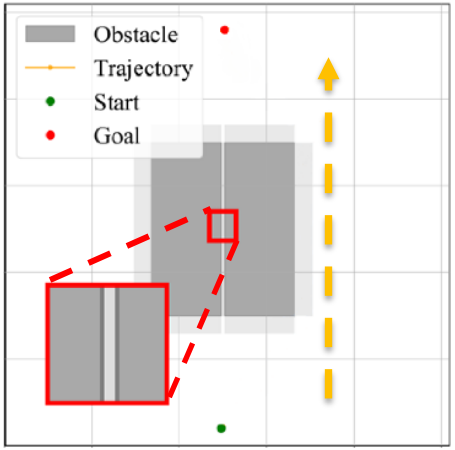


$\mathbb{E}_{p,\pi}\left[\sum_{t=0}^{\infty}\gamma^t r(s_t,a_t)\right]$: expected cumulative reward

$\pi_\theta(a|s)$: policy π parameterized with θ

Preliminary Experiments

2D Navigation tasks



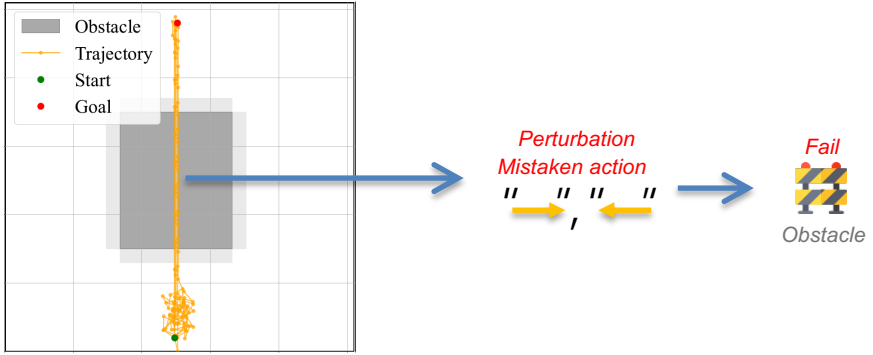
Easy task if it **'only'** goes up "↑"

What if the action is mistaken?

"✗" "→"

Preliminary Experiments

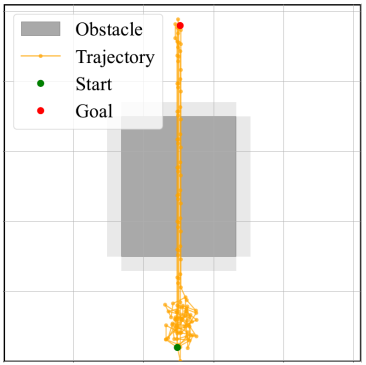
2D Navigation tasks



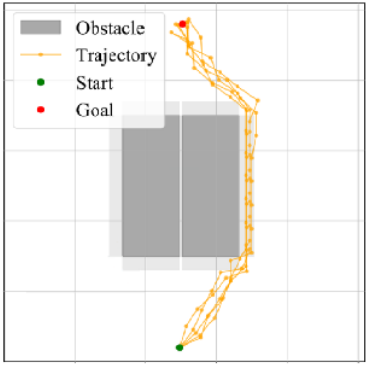
Traditional RL (PPO)

Preliminary Experiments

2D Navigation tasks



Traditional RL (PPO)



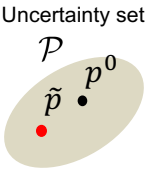
Flat reward RL (SAM+PPO)

⇒ **Flat reward RL maintains safer margins, demonstrating action robustness**

Preliminary Experiments

Goal

- Maintaining performance despite uncertainties
  in the environment: Action, Transition probability, Reward function uncertainty



Approach

- Optimizes a max-min objective to handle worst-case scenarios:
  Maximizing the return in the worst-case scenario under the Uncertainty set

Objective function

$$\max_{\pi} \min_{p \in \mathcal{P}} \mathbb{E}_{p,\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right]$$

Theoretically: covering $\mathcal{P}$

$$\max_{\pi} \min_{\|\delta_t\| \leq \beta} \mathbb{E}_{p,\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t + \delta_t) \right]$$

Practically: perturb actions + optimize

Deep Policy with Flat Reward

Objective function

$$\max_{\pi} \min_{p \in \mathcal{P}} \mathbb{E}_{p,\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right]$$

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Practically: perturb actions + optimize

Limitations

- **Impractical Assumptions**: Requires prior knowledge of uncertainty sets, unrealistic in real-world scenarios
- **Limited Scalability**: Struggles in continuous and high-dimensional environments due to the complexity of uncertainty sets and optimization
- **High Computational Cost**: Modeling uncertainties requires solving complex max-min problems, leading to significant computational overhead

Deep Policy with Flat Reward

Objective function

$$\max_{\pi} \min_{p \in \mathcal{P}} \mathbb{E}_{p,\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right]$$

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Practically: perturb actions + optimize

Our Approach

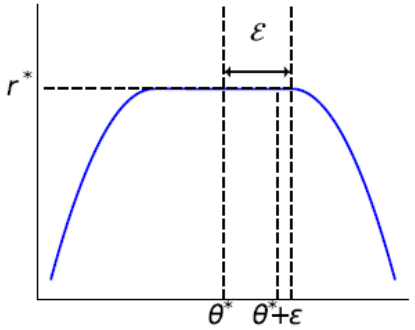
- *Rather than solving the perturbed actions, simply search for flatter reward maxima!*
- *We hope to link the flatter maxima to the action robustness.*

$$\min_{\theta} \max_{\|\epsilon\| \leq \rho} \mathbb{E}_{p,\pi_{\theta+\epsilon}} \left[\sum_{t=0}^{\infty} -\gamma^t r(s_t, a_t) \right]$$

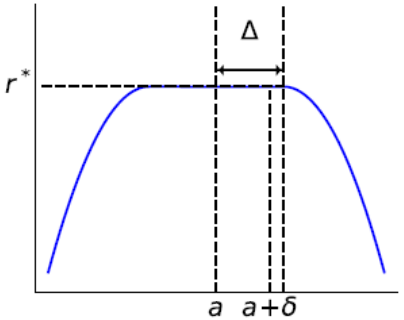
$$\max_{\pi} \min_{\|\delta_t\| \leq \beta} \mathbb{E}_{p,\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t + \delta_t) \right]$$

Deep Policy with Flat Reward

Define the reward flatness and the action robustness:



(a) $\mathcal{E}$-flat reward maxima



(b) Δ-action robust policy

Deep Policy with Flat Reward

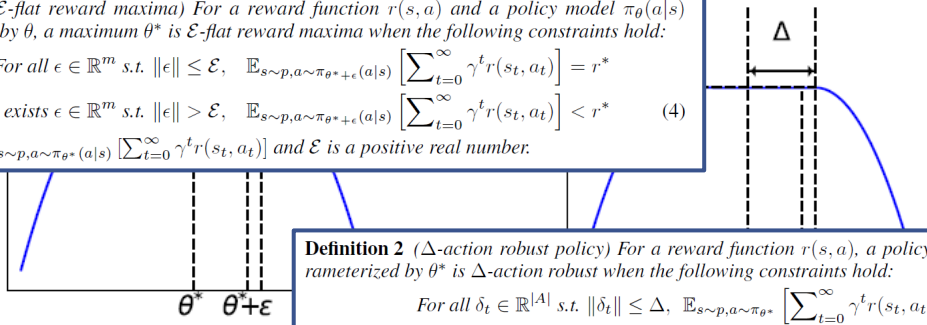
Define the reward flatness and the action robustness:

Definition 1 *($\mathcal{E}$-flat reward maxima) For a reward function $r(s,a)$ and a policy model $\pi_\theta(a|s)$ parameterized by θ, a maximum θ^* is $\mathcal{E}$-flat reward maxima when the following constraints hold:*

$$\text{For all } \epsilon \in \mathbb{R}^m \text{ s.t. } \|\epsilon\| \leq \mathcal{E}, \quad \mathbb{E}_{s\sim p, a\sim\pi_{\theta^*+\epsilon}(a|s)}\left[\sum_{t=0}^{\infty}\gamma^t r(s_t,a_t)\right] = r^*$$

$$\text{There exists } \epsilon \in \mathbb{R}^m \text{ s.t. } \|\epsilon\| > \mathcal{E}, \quad \mathbb{E}_{s\sim p, a\sim\pi_{\theta^*+\epsilon}(a|s)}\left[\sum_{t=0}^{\infty}\gamma^t r(s_t,a_t)\right] < r^* \qquad (4)$$

where $r^* := \mathbb{E}_{s\sim p, a\sim\pi_{\theta^*}(a|s)}\left[\sum_{t=0}^{\infty}\gamma^t r(s_t,a_t)\right]$ and $\mathcal{E}$ is a positive real number.



(a) $\mathcal{E}$-flat reward m

Definition 2 *(Δ-action robust policy) For a reward function $r(s,a)$, a policy model $\pi_{\theta^*}(a|s)$ parameterized by θ^* is Δ-action robust when the following constraints hold:*

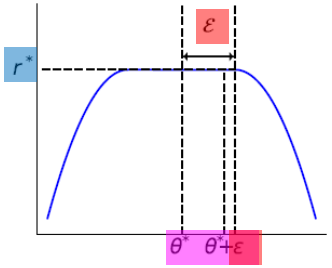
$$\text{For all } \delta_t \in \mathbb{R}^{|A|} \text{ s.t. } \|\delta_t\| \leq \Delta, \quad \mathbb{E}_{s\sim p, a\sim\pi_{\theta^*}}\left[\sum_{t=0}^{\infty}\gamma^t r(s_t,a_t+\delta_t)\right] = r^*$$

$$\text{There exists } \delta_t \in \mathbb{R}^{|A|} \text{ s.t. } \|\delta_t\| > \Delta, \quad \mathbb{E}_{s\sim p, a\sim\pi_{\theta^*}}\left[\sum_{t=0}^{\infty}\gamma^t r(s_t,a_t+\delta_t)\right] < r^*, \qquad (5)$$

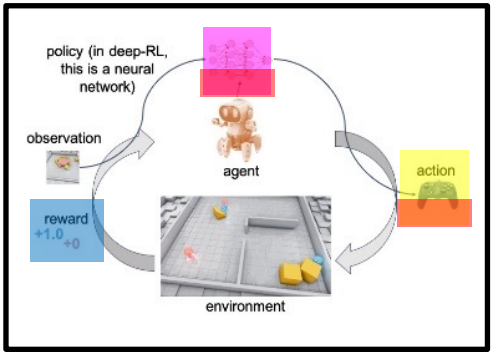
where $r^* := \mathbb{E}_{s\sim p, a\sim\pi_{\theta^*}}\left[\sum_{t=0}^{\infty}\gamma^t r(s_t,a_t)\right]$ and Δ is a positive real number.

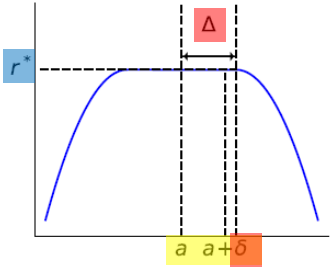
Deep Policy with Flat Reward

Connect these two definitions:



(a) $\mathcal{E}$-flat reward maxima





(b) Δ-action robust policy

Deep Policy with Flat Reward

Our mathematical claim for the link:

Proposition 1 *(Flat reward links to action robustness) If θ^* is an $\mathcal{E}$-flat reward maximum, then the policy π_{θ^*} is Δ^*-action robust, where:*

$$\Delta^* \leq \|J(\theta^*)\|\mathcal{E} + \mathcal{O}(\mathcal{E}^2), \tag{6}$$

and $J(\theta^*) := \nabla_\theta \mu_\theta(s)\big|_{\theta=\theta^*}$ is the Jacobian matrix of the mean action $\mu_\theta(s)$ with respect to θ, evaluated at θ^*.

Remark 1.1 ***(A link to Max-Min problem of action robustness)*** *For Δ^*-action robust policy derived by $\mathcal{E}$-flat reward maxima θ^*, the policy directly satisfies the objective of action robust MDP:*

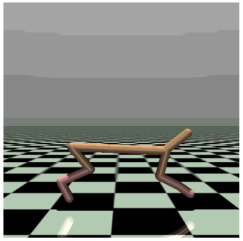
$$\theta^* = \arg\max_\theta \min_{\|\delta_t\| \leq \Delta^*} \mathbb{E}_{s\sim p, a\sim\pi_\theta}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t + \delta_t)\right], \tag{7}$$

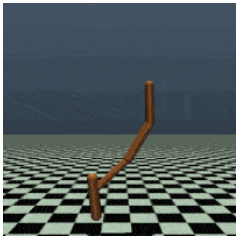
which implies that flatter reward yields the robustness against action perturbations.

Empirical Results

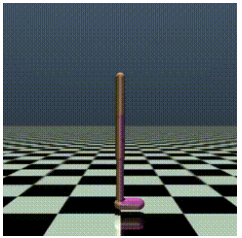
MuJoCo tasks: *Half Cheetah, Hopper, Walker2D*

Baseline comparison: Traditional RL : Proximal Policy Optimization (PPO), Robust RL, Robust Natural Actor-Critic (RNAC) (Zhou et al., 2024), Robust Adversarial Reinforcement Learning (RARL) (Pinto et al., 2017).





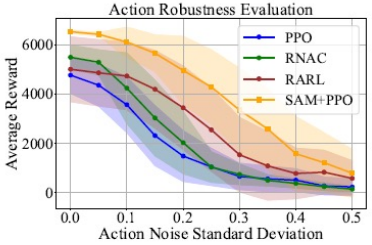
Hopper



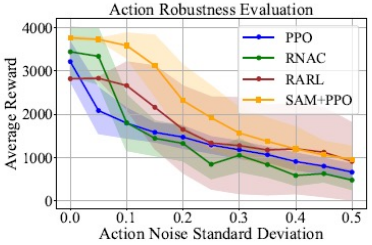
Walker2d

Empirical Results: Action Robustness

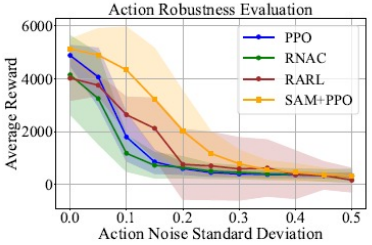
Action perturbation : added zero mean Gaussian noise



(a) HalfCheetah-v3



(b) Hopper-v3

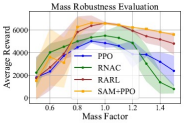


(c) Walker2d-v3

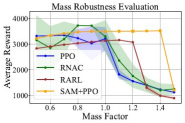
⇒ **flat reward achieved by SAM+PPO makes the policy less sensitive to action perturbations**

Empirical Results: Transition Probability

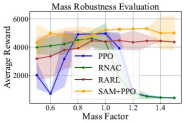
Variation in Torso Mass and Friction Coefficient



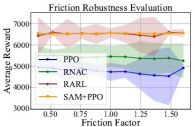
(a) HalfCheetah-v3



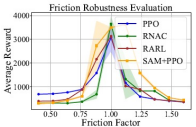
(b) Hopper-v3



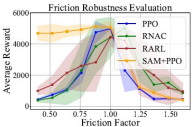
(c) Walker2d-v3



(a) HalfCheetah-v3

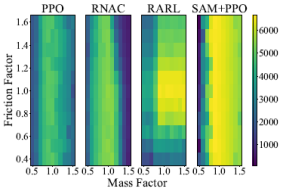


(b) Hopper-v3

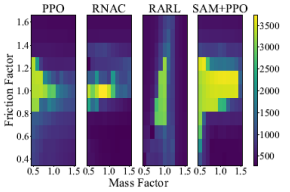


(c) Walker2d-v3

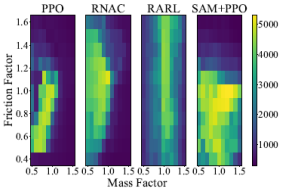
Mass and Friction Joint Variations



(a) HalfCheetah-v3



(b) Hopper-v3



(c) Walker2d-v3

Empirical Results: Noisy Reward

Reward perturbation : added Gaussian noise when training

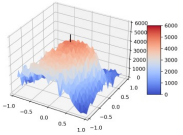
Table 2: Performance comparison of agents trained with and without reward noise ($\sigma_r = 0.1$)

Algorithm	HalfCheetah-v3		Hopper-v3		Walker2d-v3	
	Nominal	Noisy	Nominal	Noisy	Nominal	Noisy
PPO	4820	3688(−1132)	3150	2945(−205)	4780	2204(−2576)
RNAC	5423	4088(−1335)	3211	3035(−176)	4184	3172(−1012)
RARL	5620	4617(−1003)	3124	2993(−131)	4388	3085(−1303)
SAM+PPO	**6530**	**5990(−540)**	**3505**	**3377(−128)**	**5120**	**4226(−894)**

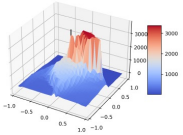
(−) values means the performance degradation from 'Nominal' to 'Noisy.'

Empirical Results: Reward Surface

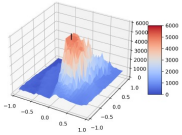
Reward perturbation : added Gaussian noise when training



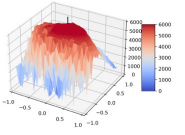
(a) PPO - HalfCheetah-v3



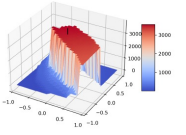
(b) PPO - Hopper-v3



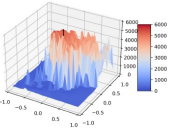
(c) PPO - Walker-v3



(d) SAM+PPO - HalfCheetah-v3



(e) SAM+PPO - Hopper-v3



(f) SAM+PPO - Walker-v3

Table 3: Flatness metrics for PPO and SAM+PPO (↓: indicates that lower is better).

Metrics	$\lambda_{\max}$ ↓ (Keskar et al., 2017)			LPF ↓ (Bisla et al., 2022)		
Environment	**HalfCheetah-v3**	**Hopper-v3**	**Walker2d-v3**	**HalfCheetah-v3**	**Hopper-v3**	**Walker2d-v3**
PPO	15192.95	131.07	7239.59	0.0385	0.00034	0.0269
SAM+PPO	**275.93**	**80.86**	**271.91**	**0.00097**	**0.00018**	**0.00028**

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Understanding Flatness in Generative Models: Its Role and Benefits



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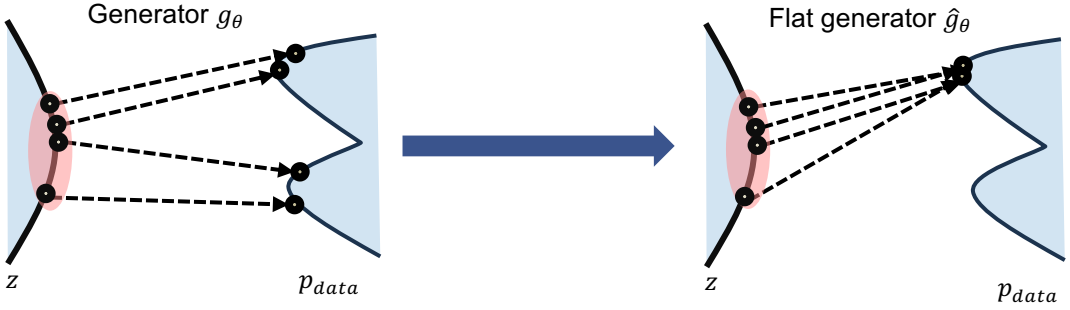


Sung Whan Yoon†
(MIIT Lab)

*Equal contribution, †Corresponding authors

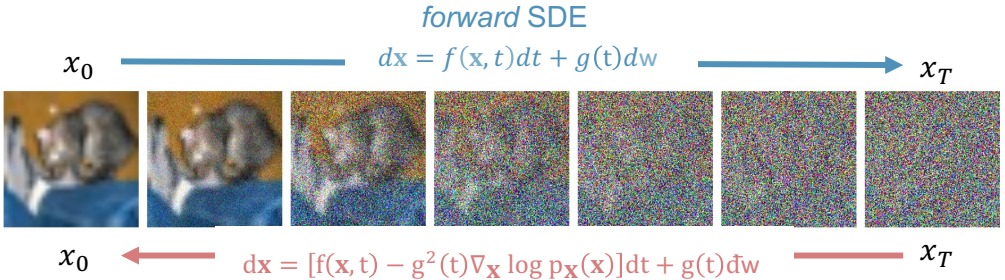
Preliminaries: Diffusion Models

Does flat generative models produce the same images when the various inputs are given?



A simple mapping from sampling to a consistent data? Maybe NOT!

Preliminaries: Diffusion Models



Score Matching Objective

$$\mathcal{L}_{SGM} = \mathbb{E}_t \left[\lambda(t) \cdot \mathbb{E}_{p_t(x)} \left[\left\| s_\theta(\mathbf{x}, t) - \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}) \right\|_2^2 \right] \right]$$

What is the Effect of Flatness in DMs?

From the motivation
"What is the role of the flatness of the Generative model?",
We **address the question and answer with Theorems 1 & 2**.

If the Diffusion Model is flat,

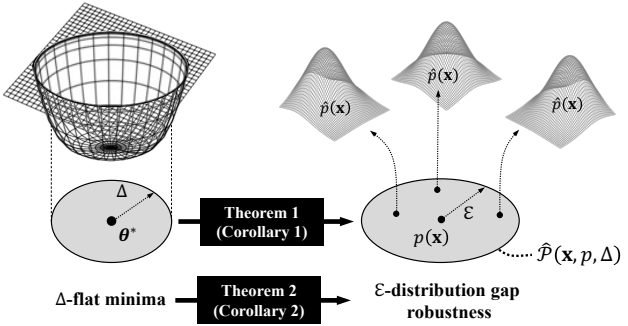
- Q1: What does the perturbed model represent in distribution?
- Q2: How robust is it to the distribution gap?

Flatness in Diffusion Models

Theorem 1. A perturbed distribution

For a given prior distribution of $p(\mathbf{x})$ and the $\boldsymbol{\delta}$-perturbed minimum, i.e., $\boldsymbol{\theta} + \boldsymbol{\delta}$, the following $\hat{p}(\mathbf{x})$ satisfies the equality:

$$\hat{p}(\mathbf{x}) = \exp\big(-I(\mathbf{x}, \boldsymbol{\delta})\big)\, p(\mathbf{x})$$



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Q1: What does the perturbed model represent in distribution?

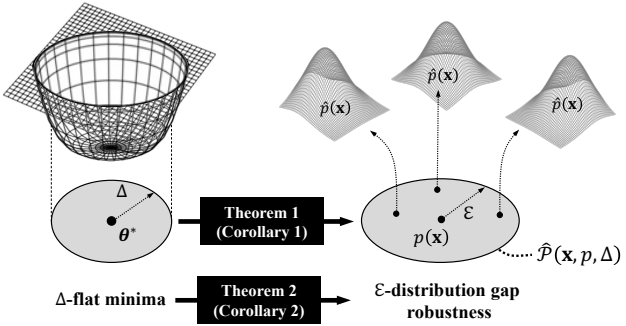
A1: Model perturbation is translated to a perturbed distribution, *resulting in a model being trained on various distributions (i.e., scaled-version of priors)*

The Effect of Flatness in Distribution Gap

Theorem 2. Link from flatness to distribution gap

A δ-flat minimum achieves $\mathcal{E}$-distribution gap robustness, such that $\mathcal{E}$ is upper-bounded as follows:

$$\mathcal{E} \leq \max_{\hat{p} \sim \hat{\mathcal{P}}(\mathbf{x}; p, \Delta)} D(p||\hat{p}).$$



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A δ-flat minimum achieves $\mathcal{E}$-distribution gap robustness, such that $\mathcal{E}$ is upper-bounded as follows:

$$\mathcal{E} \leq \max_{\hat{p} \sim \hat{\mathcal{P}}(\mathbf{x};p,\Delta)} D(p||\hat{p}).$$

Q2: How robust is it to the distribution gap?

A2: As the model has flatter minima (i.e., larger Δ), the model is robust to the distribution gap caused by Δ.

Experimental Results: Baselines

Explicit method [SAM'21]

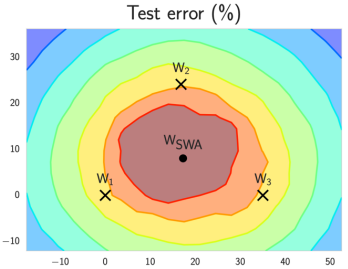
- SAM adopts the sharpness in the optimization objective [SAM'21]

$$\left[\max_{\|\epsilon\|_2 \le \rho} \mathcal{L}(w+\epsilon) - \mathcal{L}(w)\right] + \mathcal{L}(w) + h(\|w\|_2^2/\rho^2)$$

Sharpness Loss at minima L2 Reg.

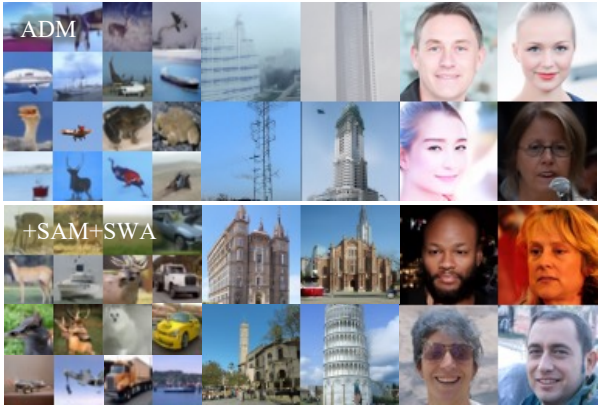
Implicit method [SWA'18, EMA'24]

- Averaging model parameters leads flat minima.
- W_1, W_2, W_3 : trained model with SGD.
- W_{SWA} : Averaged model of W_1, W_2, W_3.
- Finding flat minima results in better performance.



[SAM'21] P. Foret et al., "SHARPNESS-AWARE MINIMIZATION FOR EFFICIENTLY IMPROVING GENERALIZATION," ICLR 2021.
[EMA'24] Li, Siyuan, et al. "Switch ema: A free lunch for better flatness and sharpness." arXiv, 2024.
[SWA'18] Izmailov, Pavel, et al. "Averaging weights leads to wider optima and better generalization." UAI, 2018.

Experimental Results: Better FIDs



FID Score	CIFAR10		LSUN-Tower		FFHQ	
	T=20	T=100	T=20	T=100	T=20	T=100
ADM	34.47	8.80	36.65	8.57	30.81	7.53
+EMA	10.63	4.06	7.87	2.49	19.03	6.19
+SWA	11.00	3.78	8.72	2.31	17.93	5.49
IP	20.11	7.23	25.77	7.00	15.03	13.55
+EMA	9.10	3.46	7.66	2.43	11.72	4.00
+SWA	9.04	3.07	8.55	2.34	12.99	**3.54**
SAM	9.01	3.83	16.02	4.79	11.59	5.29
+EMA	**7.00**	3.18	6.66	2.30	**11.41**	5.04
+SWA	7.27	**2.96**	**6.50**	**2.27**	12.15	4.17

SAM (+EMA, +SWA) achieves comparable or better FID score.

Experimental Results: Robust to Low Precision

FID Score	T=20		T=100	
	32 bit	8 bit	32 bit	8 bit
ADM	34.47	48.02 (+13.65)	8.80	12.78 (+3.98)
+EMA	10.63	20.65 (+10.02)	4.06	7.36 (+3.3)
+SAM	9.01	8.94 (-0.07)	3.83	4.02 (+0.19)
+SAM+EMA	7.00	7.20 (+0.2)	3.18	3.12 (-0.06)

SAM (+EMA) shows robustness to **8-bit quantization**.
SAM raises robustness to quantization,
which is **essential for model deployment**.

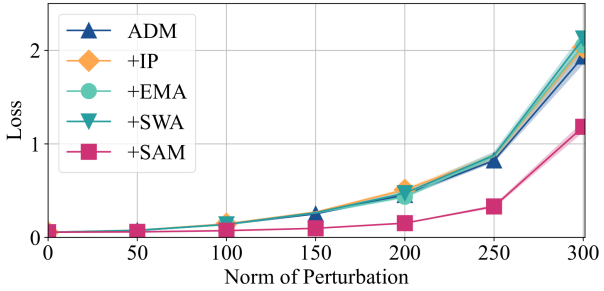
T: sampling steps

Experimental Results: Robust to Low Precision



While **ADM** and **+IP** collapse in **4-bit quantization**, **SAM** maintains the image generation performance.

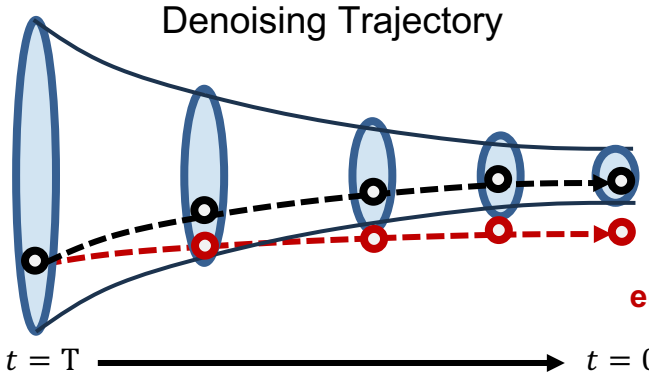
Experimental Results: Flatness



LPF ↓	w/o	+EMA	+SWA
ADM	0.097	0.099	0.099
+IP	0.103	0.101	0.102
+SAM	**0.063**	**0.063**	**0.063**

While **+SAM** finds a flatter loss landscape **explicitly**, empirical methods (**+EMA, +SWA**) shows less impact.

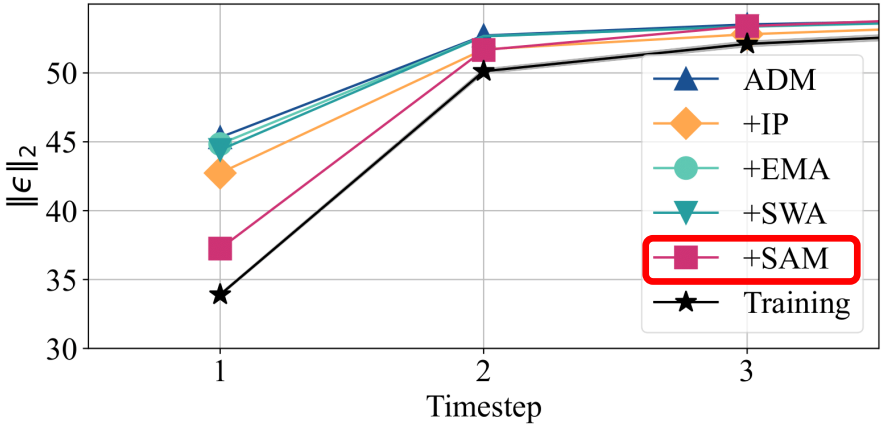
Experimental Results: Exposure Bias



Because of the **iterative process in Diffusion Models**,
the **error accumulates** in the sampling process.
The accumulation of errors is commonly referred to as exposure bias. [IP'23]

[IP'23] Ning, Mang, et al. "Input perturbation reduces exposure bias in diffusion models." ICML, 2023.

Experimental Results: Exposure Bias



Flat minima show robustness to **Exposure bias**, where **+SAM** shows closer behavior with **Training.**

Takeaway

A loss landscape perspective is a theoretical lens to understand deep models.

We theoretically extend the concept to deep reinforcement models [ICLR'25] and generative models [ICCV'25].